



**CEBU INSTITUTE OF TECHNOLOGY**  
**U N I V E R S I T Y**

# IT342- G1

## System Integration and Architecture

---

### **System Design Document (SDD)**

---

Project Title: Fort Point Properties

Prepared By: Aligato, Zander Navarro

Version: 1.2

Date: February 27, 2026

Status: Final

## REVISION HISTORY TABLE

| <b>Versio<br/>n</b> | <b>Date</b>       | <b>Author</b>  | <b>Changes Made</b>              | <b>Status</b> |
|---------------------|-------------------|----------------|----------------------------------|---------------|
| 0.1                 | February 21, 2026 | Zander Aligato | Initial draft                    | Final         |
| 0.2                 | February 26, 2026 | Zander Aligato | -                                | -             |
| 0.3                 | [Date]            | [Your Name]    | Updated database design          | Review        |
| 0.4                 | [Date]            | [Your Name]    | Added UI/UX designs              | Review        |
| 0.5                 | [Date]            | [Your Name]    | Incorporated feedback            | Revised       |
| 0.6                 | [Date]            | [Your Name]    | Final review and corrections     | Final         |
| 1                   | [Date]            | [Your Name]    | Baseline version for development | Approved      |

## TABLE OF CONTENTS

### Contents

|                                           |    |
|-------------------------------------------|----|
| EXECUTIVE SUMMARY                         | 4  |
| 1.0 INTRODUCTION                          | 5  |
| 2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION | 5  |
| 3.0 NON-FUNCTIONAL REQUIREMENTS           | 8  |
| 4.0 SYSTEM ARCHITECTURE                   | 9  |
| 5.0 API CONTRACT & COMMUNICATION          | 11 |
| 6.0 DATABASE DESIGN                       | 13 |
| 7.0 UI/UX DESIGN                          | 14 |
| 8.0 PLAN                                  | 17 |

# EXECUTIVE SUMMARY

## 1.1 Project Overview

Fort Point Properties is a comprehensive real estate platform designed to connect property buyers with premium developer-based project listings. The platform functions as a unified ecosystem where buyers can discover properties, agents can access detailed listing information, and administrators can manage all content and team operations.

## 1.2 Objectives

1. Provide a centralized real estate ecosystem that bridges the gap between property seekers, sales professionals, and administrators.
2. Equip real estate agents with a collaborative platform featuring detailed property information and client engagement tools.
3. Deliver a seamless experience for registered buyers to access advanced site features, expanded property details, and direct communication tools.
4. Provide administrators with tools to oversee content management and operational workflows.

## 1.3 Scope

### Included Features:

- Property listing management with developer-based project details (admin-controlled)
- Property listing with search functionality
- User registration provides enhanced access to in-app messaging, detailed listings, and job applications.
- In-app messaging for direct communication between buyers and agents.
- Activity tracking, allowing users to save "Favorite" property listings
- Agent portal with dashboards for properties with detailed project details, "pitch-ready" section, and developer inventory links.
- End-to-end career and recruitment portal for job applicants and admin reviews.
- Web Interface and Native Mobile Application
- MySQL database with relational schema

**Excluded Features:**

- AI-powered chatbots.
- Admin Sales monitoring and performance tracking.
- Agent training and learning modules.
- Direct payment processing gateways.

## 1.0 INTRODUCTION

### 1.1 Purpose

This document serves as the design specification for the Fort Point Properties platform. It provides detailed requirements, functional specifications, user journeys, and acceptance criteria to guide the development of the real estate ecosystem.

## 2.0 FUNCTIONAL REQUIREMENTS SPECIFICATION

### 2.1 Project Overview

**Project Name:** Fort Point Properties

**Domain:** Real Estate Platform

**Primary Users:** Internal Users (Agents/Admins) and External Users (Buyers/Applicants)

**Problem Statement:** Property seekers need a streamlined way to find developer-based projects while sales professionals and administrators require centralized tools to manage listings, leads, and operational workflows.

**Solution:** A unified platform focused on developer-based projects that transforms the property search process into an efficient business management and lead-generation engine.

### 2.2 Core User Journeys

#### Journey 1: First-Time / Guest Users

1. User visits the homepage
2. Sees an initial preview of basic property listings
3. Browse general information including "About Us" and "Contact" details.
4. Read market news articles.

5. Click "Sign Up" to register for a full account.

### **Journey 2: Registered Users**

1. User logs into the authenticated homepage.
2. Browses the platform and views the full, expanded preview of a property listing.
3. Saves preferred properties to their "Favorites" list.
4. Uses the standard (non-AI) in-app chatbox to communicate directly with an assigned agent
5. Navigates to the careers page and submits an application for a job/career opportunity.

### **Journey 3: Agent Users**

1. Agent logs into the specialized agent portal
2. Views admin-created listings populated with full "pitch-ready" data and amenity badges.
3. Accesses developer resources via one-click inventory links for client presentations
4. Checks the internal bulletin board
5. Responds to in-app messages sent by registered property buyers.

### **Journey 4: Administrator Management**

1. Admin logs in with full system administration access.
2. Navigates to property inventory to create or edit listings
3. Formats unit types using a structured unit configuration grid and uploads developer resources (PDFs/Links).
4. Publishes market articles
5. Reviews incoming career applications via the admin dashboard, updating statuses (pending/approved/rejected)

## **2.3 Feature List (MoSCoW)**

### **MUST HAVE**

1. User registration and authentication.
2. Basic and expanded property listing views.
3. In-app messaging (standard chatbox).
4. Admin property management (create/edit listings)
5. Agent portal with access developer resources
6. Logout Functionality

### **SHOULD HAVE**

1. Job application workflow and admin review dashboard

2. Activity tracking and "Favorite" listing capabilities.
3. Articles and Internal Bulletin Boards

### **COULD HAVE**

1. User Profile Management
2. Property Listing Filters
3. Interactive Amortization Calculator
4. Side-by-Side Property Comparison Matrix
5. Neighborhood / Proximity Map

### **WON'T HAVE**

1. AI-powered chat capabilities.
2. Agent training modules.
3. Team performance reports.
4. Multi-language support
5. Forms Management Tab

## **2.4 Detailed Feature Specifications**

### **Feature: User Authentication & Registration**

- **Screens:** Registration Form, Login, Forgot Password
- **Fields:** First Name, Last Name, Email, Password, Confirm Password
- **Validation:** Email format, password strength, unique email verification
- **API Endpoints:** POST /auth/register, POST /auth/login, POST /auth/logout
- **Security:** JWT tokens for session management, bcrypt password hashing
- **Process:** Converts unauthenticated Public Users into Registered Users, unlocking expanded site features.

### **Feature: Property Catalog**

- **Screens:** Admin Property Listing (CRUD), Agent Property Listing Detailed Preview (Read), Authenticated User Property Browsing, Guest User Property Browsing
- **Fields:** Property Name, Description, Developer Name, Price Range, Location, Description, Property / Unit Type, Listing Type (Pre-Selling, Ready-For-Occupancy), Pet Friendly, Parking Availability, Turn Over Date, Amenities, Price Computations, Developer Links / PDFs, Pitch-Ready Phrases.
- **Display:** Form, List, Card Displays
- **Admin Functions:** Create, read, update, and delete (CRUD) all property listings. Upload developer resources such as PDFs and external links.

- **Access Control:** Restricted to Admin users exclusively.
- **API Endpoints:** POST /admin/properties, PUT /admin/properties/{id}, DELETE /admin/properties/{id}

### Feature: Property Browsing & User Engagement

- **Screens:** Public Homepage (Basic Preview), Authenticated Homepage (Expanded Property Detail), Agent Property (with Pitch Ready Section and Developer Links / PDFs)
- **Display:** Expanded Window with Advance detailed property specifications visible only to Registered Users and above
- **Functions:** Standard (non-AI) in-app messaging to communicate directly with assigned agents. Ability to save "Favorite" listings
- **Persistence:** Database storage per user to track inquiry history and saved favorites.
- **API Endpoints:** GET /properties (Basic), GET /properties/{id}/advanced (Registered/Agent/Admin), POST /messages, POST /users/favorites, GET /users/favorites, DELETE /users/favorites/id, GET /agent/messages

### Feature: Agent Information Portal

- **Screens:** Agent Portal Dashboard, Internal Bulletin Board, Prospect In-app Messaging
- **Display:** Admin-created listings formatted with "pitch-ready" data and property browsing
- **Functions:** One-click access to developer inventory links. Ability to read and respond to direct messages from Registered Users.
- **Access Control:** Restricted to Agent and Admin roles
- **API Endpoints:** GET /agent/dashboard, POST /agent/messages/reply

### Feature: Application & Recruitment Portal

- **Screens:** Public Careers Page, Application Submission Form, Admin Review, Agent Account Creation.
- **Data Collected:** Applicant contact information, and resume uploads.
- **Process:** End-to-end job application workflow including submission and status tracking (pending, approved, or rejected).
- **Admin Functions:** Review and update the status of submitted career applications.
- **API Endpoints:** POST /careers/apply, GET /admin/applications, PUT /admin/applications/{id}/status



## **2.5 Acceptance Criteria**

### **AC-1: Successful User Registration and Access**

- Given I am a First-Time/Guest User browsing the basic listings
- When I navigate to the sign-up page and enter valid registration details
- And I submit the form
- Then my account should be successfully created
- And I should be granted access to the authenticated user homepage and expanded property views

### **AC-2: Registered User Inquiry via Chatbox**

- Given I am logged in as a Registered Property Buyer
- When I view the full, expanded preview of a listing
- And I submit an inquiry using the direct in-app messaging chatbox
- Then the message should be routed directly to an agent
- And the inquiry should be logged in my personal activity tracking history

### **AC-3: Admin Product Management**

- Given I am logged in as a Real Estate Office Administrator
- When I create a new listing in the property
- And I populate the unit types using the standardized (real estate) property breakdown format
- And I upload developer resources such as PDFs or links
- Then the basic listing details should become immediately visible to Guest
- And the full "pitch-ready" data and amenity badges should become available in the Agent Portal

### **AC-4: Job Application Submission and Tracking**

- Given I am a Prospective Agent on the public careers
- When I complete and submit a career application
- Then my application status should be recorded as "pending"
- And the application should appear in the Admin review dashboard for processing

## **3.0 NON-FUNCTIONAL REQUIREMENTS**

### **3.1 Performance Requirements**

- The web and mobile applications should load pages within 3 to 5 seconds on a standard internet connection.

- The system must handle at least 50 simultaneous users without crashing.
- Database query execution for property catalog search must not exceed 500ms.

### 3.2 Security Requirements

- HTTPS for all communications
- Cross-Origin Resource Sharing (CORS)
- JWT token authentication
- Password hashing with bcrypt (salt rounds = 12)
- SQL injection prevention
- XSS protection
- Rate limiting: 100 requests/minute per IP
- Admin endpoints require role verification

### 3.3 Compatibility Requirements

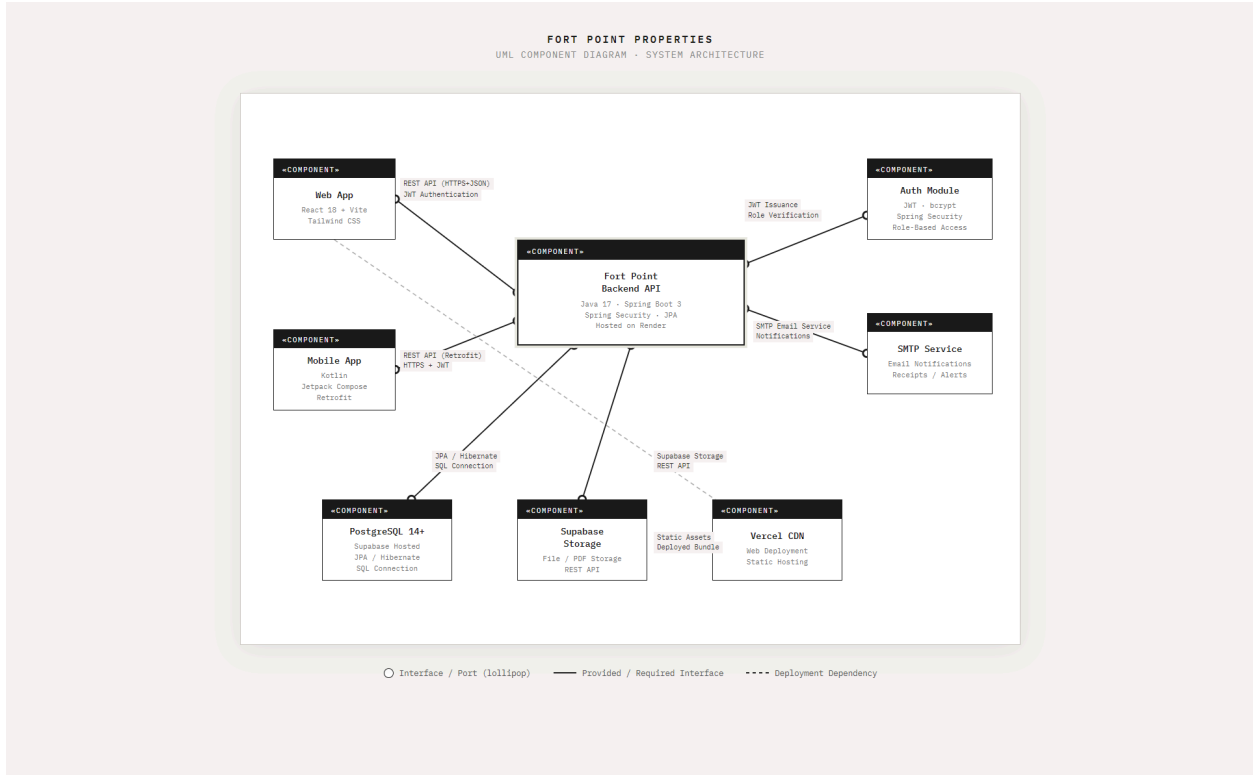
- **Web Browsers:** Chrome, Firefox, Safari, Edge (latest 2 versions)
- **Android:** API Level 24+ (Android 7.0+)
- **Screen Sizes:** Mobile (360px+), Desktop (1024px+)
- **Operating Systems:** Windows 10+

### 3.4 Usability Requirements

- WCAG 2.1 Level AA compliance for web
- Consistent navigation across public registered, agent, and admin interfaces.
- Touch targets minimum 44x44px on mobile

## 4.0 SYSTEM ARCHITECTURE

### 4.1 Component Diagram



## Technology Stack:

- **Backend:** Java 17, Spring Boot 3.5.11, Spring Security, and Spring Data JPA.
- **Database:** PostgreSQL hosted in Supabase
- **Web Frontend:** React 18 Vite, Tailwind CSS
- **Mobile:** Kotlin, Jetpack Compose, XML (for specific legacy layouts/resources), Retrofit
- **Deployment & Hosting:** Vercel (Frontend), Render (Backend), Supabase (Database)

## 5.0 API CONTRACT & COMMUNICATION

### 5.1 API Standards

|                           |                                                                        |
|---------------------------|------------------------------------------------------------------------|
| <b>Base URL</b>           | https://[server_hostname]:[port]/api/v1                                |
| <b>Format</b>             | JSON for all requests/responses                                        |
| <b>Authentication</b>     | Bearer token (JWT) in Authorization header                             |
| <b>Response Structure</b> | <pre>{   "success": boolean,   "data": object null,   "error": {</pre> |

|  |                                                                                                 |
|--|-------------------------------------------------------------------------------------------------|
|  | <pre> "code": string, "message": string, "details": object null }, "timestamp": string } </pre> |
|--|-------------------------------------------------------------------------------------------------|

## 5.2 Endpoint Specifications

### Authentication Endpoints

#### User Registration

|                            |                                                                                                                                                                                                                                                                                                                                                    |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>         | Registers a new public user                                                                                                                                                                                                                                                                                                                        |
| <b>API URL</b>             | /auth/register                                                                                                                                                                                                                                                                                                                                     |
| <b>HTTP Request Method</b> | POST                                                                                                                                                                                                                                                                                                                                               |
| <b>Format</b>              | JSON for all requests/responses                                                                                                                                                                                                                                                                                                                    |
| <b>Authentication</b>      | None                                                                                                                                                                                                                                                                                                                                               |
| <b>Request Payload</b>     | <pre> {   "email": "juan@example.com",   "password": "SecurePassword123!",   "firstname": "Juan",   "lastname": "Dela Cruz" } </pre>                                                                                                                                                                                                               |
| <b>Response Structure</b>  | <pre> {   "success": true,   "data": {     "user": {       "id": "usr_01",       "email": "juan@example.com",       "firstname": "Juan",       "lastname": "Dela Cruz",       "role": "registered_user"     },     "accessToken": "eyJhbG...",     "refreshToken": "dGVzd..."   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" } </pre> |

#### User Login

|                    |                                                 |
|--------------------|-------------------------------------------------|
| <b>Description</b> | Authenticates a user and returns session tokens |
| <b>API URL</b>     | /auth/login                                     |

|                           |                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HTTP Method</b>        | POST                                                                                                                                                                                                                                                                                                                                             |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                                                                                                                                                                                                                  |
| <b>Authentication</b>     | None                                                                                                                                                                                                                                                                                                                                             |
| <b>Request Payload</b>    | <pre>{   "email": "juan@example.com",   "password": "SecurePassword123!" }</pre>                                                                                                                                                                                                                                                                 |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "user": {       "id": "usr_01",       "email": "juan@example.com",       "firstname": "Juan",       "lastname": "Dela Cruz",       "role": "registered_user"     },     "accessToken": "eyJhbG...",     "refreshToken": "dGVzd..."   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |

## User Logout

|                           |                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>        | Invalidates the current user's refresh token and terminates the session.                                                                            |
| <b>API URL</b>            | /auth/logout                                                                                                                                        |
| <b>HTTP Method</b>        | POST                                                                                                                                                |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                     |
| <b>Authentication</b>     | Bearer Token (Authenticated User, Agent, Admin)                                                                                                     |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "message": "Successfully logged out."   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |

## Property & Catalog Endpoints

### Get All Properties (Public / Basic)

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>        | Retrieves basic property details for the public homepage                                                                                                                                                                                                                                                                                                                                                                              |
| <b>API URL</b>            | /properties                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>HTTP Method</b>        | GET                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Authentication</b>     | None                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Query Parameters</b>   | ?page=1&limit=10                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": [     {       "id": "prop_123",       "name": "Sadya Homes",       "developer": "ABC Developers",       "location": "Consolacion, Cebu",       "basic_description": "A quiet subdivision...",       "listing_type": "Pre-Selling"     }   ],   "meta": {     "current_page": 1,     "total_pages": 5,     "total_records": 48   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |

### Get Property Details (Advanced View)

|                           |                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------|
| <b>Description</b>        | Retrieves expanded details, including unit breakdowns, pitch-ready phrases, and developer resources |
| <b>API URL</b>            | /properties/{id}/advanced                                                                           |
| <b>HTTP Method</b>        | GET                                                                                                 |
| <b>Format</b>             | JSON for all requests/responses                                                                     |
| <b>Authentication</b>     | Bearer Token (Registered User, Agent, Admin)                                                        |
| <b>Response Structure</b> | <pre>{   "success": true,</pre>                                                                     |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <pre> "data": {   "id": "prop_123",   "name": "Sadya Homes",   "developer": "ABC Developers",   "amenities": ["Pool", "Clubhouse"],   "developer_resources": {     "brochure_pdf": "https://pdf.url",     "inventory_link": "https://dev.url"   },   "units": [     {       "unit_type": "Studio",       "equity_per_month": 5000.00,       "bank_financing": 1500000.00     }   ] }, "error": null, "timestamp": "2026-02-28T09:47:13Z" } </pre> |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Admin Create Property

|                        |                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | Creates a new property listing with detailed unit breakdowns and resources                                                                                                                                                                                                                                                                                                             |
| <b>API URL</b>         | /admin/properties                                                                                                                                                                                                                                                                                                                                                                      |
| <b>HTTP Method</b>     | POST                                                                                                                                                                                                                                                                                                                                                                                   |
| <b>Format</b>          | JSON for all requests/responses                                                                                                                                                                                                                                                                                                                                                        |
| <b>Authentication</b>  | Bearer Token (Admin Role Required)                                                                                                                                                                                                                                                                                                                                                     |
| <b>Request Payload</b> | <pre> {   "name": "Sadya Homes",   "developer": "ABC Developers",   "location": "Consolacion, Cebu",   "listing_type": "Pre-Selling",   "basic_description": "A quiet subdivision...",   "pitch_ready_data": {     "key_selling_points": [       "Near SM Consolacion",       "Flood-free area",       "24/7 security"     ],   }, } </pre> <p>"amenities": ["Pool", "Clubhouse"],</p> |

|                           |                                                                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | <pre> "developer_resources": {   "brochure_pdf": "https://pdf.url",   "inventory_link": "https://dev.url" },  "units": [   {     "unit_type": "Studio",     "equity_per_month": 5000.00,     "bank_financing": 1500000.00   } ] } </pre> |
| <b>Response Structure</b> | <pre> {   "success": true,   "data": {     "message": "Property created successfully.",     "property_id": "prop_124"   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" } </pre>                                               |

## In-App Messaging

### Inquiry Messaging

|                        |                                                                                                                                                                                       |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>     | Creates a new inquiry regarding a specific property. The inquiry is broadcast to all available agents. The first agent to reply will be assigned and the conversation will be locked. |
| <b>API URL</b>         | /conversations                                                                                                                                                                        |
| <b>HTTP Method</b>     | POST                                                                                                                                                                                  |
| <b>Format</b>          | JSON for all requests/responses                                                                                                                                                       |
| <b>Authentication</b>  | Bearer (Registered User)                                                                                                                                                              |
| <b>Request Payload</b> | <pre> {   "property_id": "prop_123",   "message_body": "Is the Studio unit available?" } </pre>                                                                                       |



|                           |                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "conversation_id": "conv_789",     "status": "waiting_for_agent",     "message": "Your inquiry has been sent to available agents."   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Agent Reply to Inquiry

|                           |                                                                                                                                                                                                                                                                                                                                             |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>        | Allows an agent to reply to an unassigned inquiry. The first agent who replies becomes permanently assigned to the conversation.                                                                                                                                                                                                            |
| <b>API URL</b>            | /conversations/{conversation_id}/claim                                                                                                                                                                                                                                                                                                      |
| <b>HTTP Method</b>        | POST                                                                                                                                                                                                                                                                                                                                        |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                                                                                                                                                                                                             |
| <b>Authentication</b>     | Bearer (Agent Only)                                                                                                                                                                                                                                                                                                                         |
| <b>Request Payload</b>    | <pre>{   "message_body": "Yes, the Studio unit is still available." }</pre>                                                                                                                                                                                                                                                                 |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "conversation_id": "conv_789",     "assigned_agent_id": "usr_456",     "status": "active"   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> <p><i>//if already claimed</i></p> <pre>{   "success": false,   "data": null,   "error": {     "code": "CONVERSATION_LOCKED",</pre> |

|  |                                                                                                                           |
|--|---------------------------------------------------------------------------------------------------------------------------|
|  | <pre>"message": "This inquiry has already been assigned to another agent." }, "timestamp": "2026-02-28T09:47:13Z" }</pre> |
|--|---------------------------------------------------------------------------------------------------------------------------|

## Agent to Buyer Conversation

|                           |                                                                                                                                                                                                     |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Description</b>        | Sends a message within an already active conversation between buyer and assigned agent.                                                                                                             |
| <b>API URL</b>            | /conversations/{conversation_id}/messages                                                                                                                                                           |
| <b>HTTP Method</b>        | POST                                                                                                                                                                                                |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                                                                     |
| <b>Authentication</b>     | Bearer (Registered User or Assigned Agent)                                                                                                                                                          |
| <b>Request Payload</b>    | <pre>{   "message_body": "Can I schedule a site visit?" }</pre>                                                                                                                                     |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "message_id": "msg_890",     "conversation_id": "conv_789",     "status": "delivered"   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |

## Career Applications

|                    |                                                              |
|--------------------|--------------------------------------------------------------|
| <b>Description</b> | Submits a new job application to the admin review dashboard. |
|--------------------|--------------------------------------------------------------|

|                           |                                                                                                                                                                                                             |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>API URL</b>            | /careers/apply                                                                                                                                                                                              |
| <b>HTTP Method</b>        | POST                                                                                                                                                                                                        |
| <b>Format</b>             | JSON for all requests/responses                                                                                                                                                                             |
| <b>Authentication</b>     | Bearer (Registered User)                                                                                                                                                                                    |
| <b>Request Payload</b>    | <pre>{   "phone_number": "+639123456789",   "resume_link":     "https://storage.url/resume.pdf",   "cover_letter": "I have 5 years of     experience..." }</pre>                                            |
| <b>Response Structure</b> | <pre>{   "success": true,   "data": {     "application_id": "app_001",     "status": "pending",     "message": "Application submitted."   },   "error": null,   "timestamp": "2026-02-28T09:47:13Z" }</pre> |

## 5.3 Error Handling

### HTTP Status Codes

- **400 Bad Request:** The request was malformed, or the payload failed validation (e.g., missing required fields like email or password).
- **401 Unauthorized:** Authentication failed. The user did not provide a valid JWT, or the token has expired.
- **403 Forbidden:** The user is authenticated but lacks the necessary permissions to access the endpoint (e.g., a Registered Buyer attempting to access an Admin-only `/admin/properties` route).
- **404 Not Found:** The requested resource does not exist (e.g., fetching a property ID that was deleted).
- **409 Conflict:** The request conflicts with the current state of the server (e.g., attempting to register with an email address that is already in use).
- **429 Too Many Requests:** The user has exceeded the rate limit of 100 requests/minute.
- **500 Internal Server Error:** An unexpected condition was encountered on the server.

## Error Code Examples

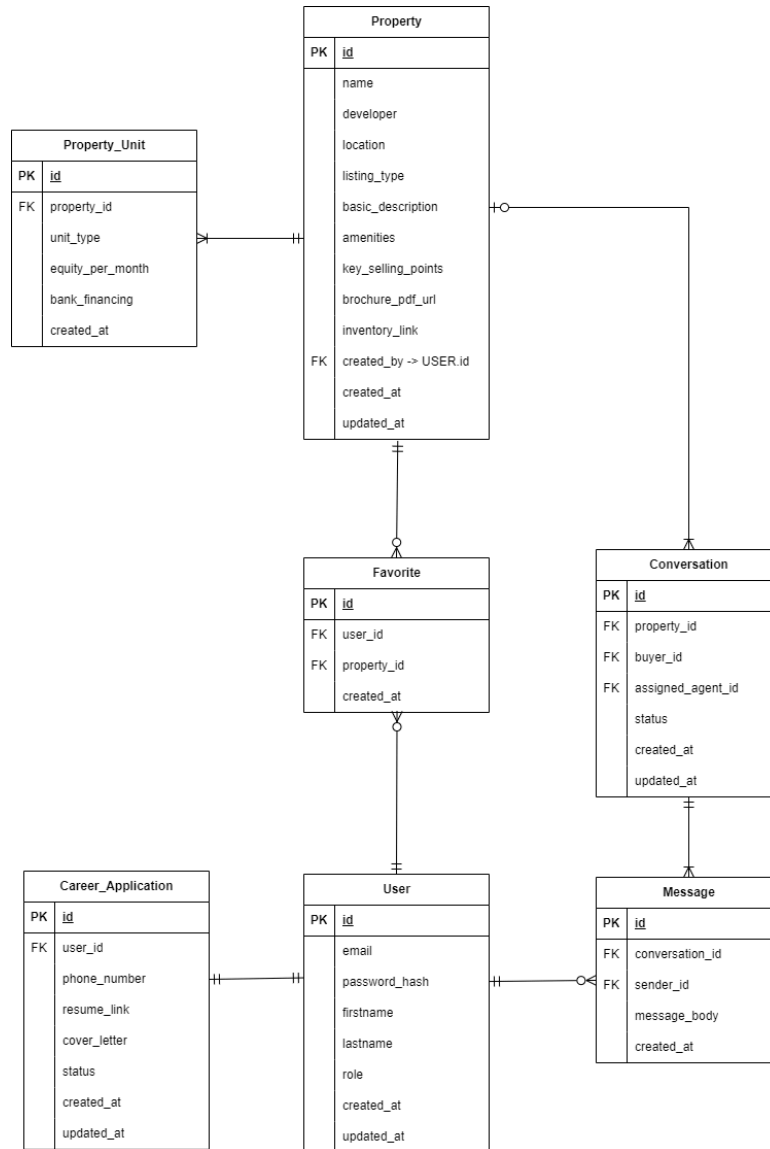
|           |                                                                                                                                                                                                                                                                       |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Example 1 | <pre>{   "success": false,   "data": null,   "error": {     "code": "PROP-001",     "message": "The selected unit is no longer available.",     "details": "Unit Studio in Sadya Homes has already been reserved."   },   "timestamp": "2026-02-28T09:47:13Z" }</pre> |
| Example 2 | <pre>{   "success": false,   "data": null,   "error": {     "code": "VALID-002",     "message": "Invalid file format.",     "details": "Only PDF and DOCX files are allowed."   },   "timestamp": "2026-02-28T09:47:13Z" }</pre>                                      |

## Common Error Codes

- **AUTH-001:** Invalid email or password.
- **AUTH-002:** JWT token missing, invalid, or expired.
- **AUTH-003:** Insufficient role permissions (Forbidden).
- **VALID-001:** Payload validation failed (missing or incorrect data format).
- **VALID-002:** Invalid file format or size exceeded
- **CHAT-001:** Conversation already assigned to another agent.
- **DB-001:** Resource not found (Property, User, or Message ID does not exist).
- **DB-002:** Duplicate entry (e.g., email already registered).
- **PROP-001:** Property or unit no longer available / Listing closed.
- **RATE-001:** Rate limit exceeded (Triggered by standard DDOS/spam protection).
- **SYSTEM-001:** Unexpected internal server error or database connection failure.

## 6.0 DATABASE DESIGN

### 6.1 Entity Relationship Diagram



### Detailed Relationships:

- **One-to-Many: User -> Favorites:** A registered user can save multiple properties to their favorites list.
- **One-to-Many: Property -> Favorites:** A single property can be favorited by multiple users.
- **One-to-Many: Property -> Units:** A single property listing contains multiple structured unit types (e.g., Studio, 1BR, 2BR).
- **One-to-Many: User -> Career\_Applications:** A prospective agent (User) can submit a job application.
- **One-to-Many: Property -> Conversations:** A property can have multiple inquiry conversations tied to it.

- **One-to-Many: User -> Conversations (as Buyer):** A registered user can start multiple inquiries.
- **One-to-Many: User -> Conversations (as Agent):** An agent can claim and be assigned to multiple inquiries.
- **One-to-Many: Conversation -> Messages:** A single claimed conversation contains multiple individual chat messages.
- **One-to-Many: User -> Messages (as Sender):** A user (buyer or agent) sends multiple messages.

#### Key Tables:

1. **users:** Stores all account information and role-based access levels (Guest/Registered, Agent, Admin).
2. **properties:** Stores the main project information, descriptions, and developer resources.
3. **property\_units:** Stores the 3-column pricing and financing data linked to specific properties.
4. **favorites:** A junction table tracking which users have saved which properties.
5. **conversations:** Manages the inquiry lifecycle (Broadcast -> Claimed) by linking a buyer, an assigned agent, and a property.
6. **messages:** Stores the individual chat lines within an active conversation.
7. **career\_applications:** Stores recruitment submissions and tracks administrative review statuses.

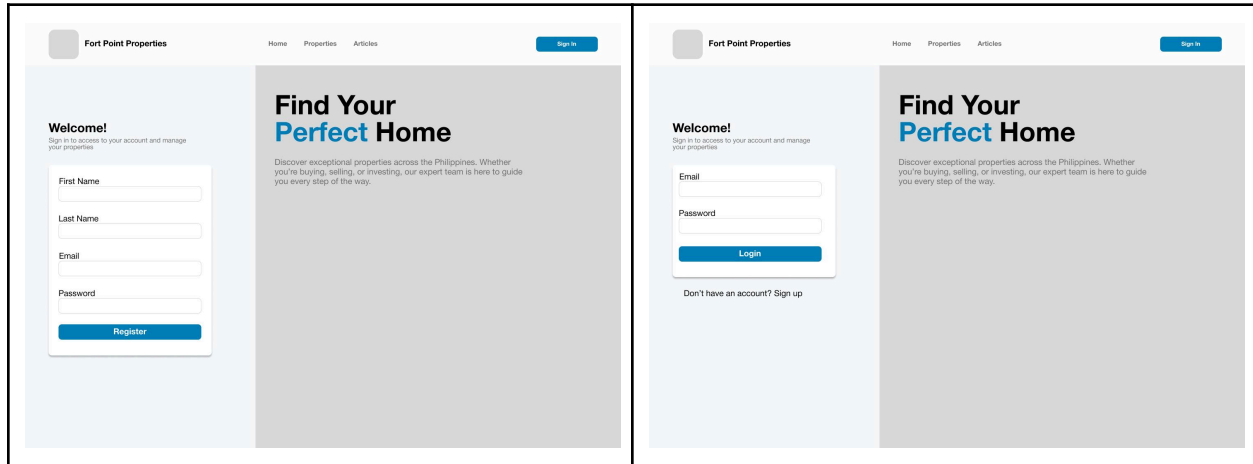
#### Table Structure Summary:

- **users:** id, email, password\_hash, firstname, lastname, role, created\_at, updated\_at
- **properties:** id, name, developer, location, listing\_type, basic\_description, amenities, key\_selling\_points, brochure\_pdf\_url, inventory\_link, created\_by, created\_at, updated\_at
- **property\_units:** id, property\_id, unit\_type, equity\_per\_month, bank\_financing, created\_at
- **favorites:** id, user\_id, property\_id, created\_at
- **conversations:** id, property\_id, buyer\_id, assigned\_agent\_id, status, created\_at, updated\_at
- **messages:** id, conversation\_id, sender\_id, message\_body, created\_at
- **career\_applications:** id, user\_id, phone\_number, resume\_link, cover\_letter, status, created\_at, updated\_at

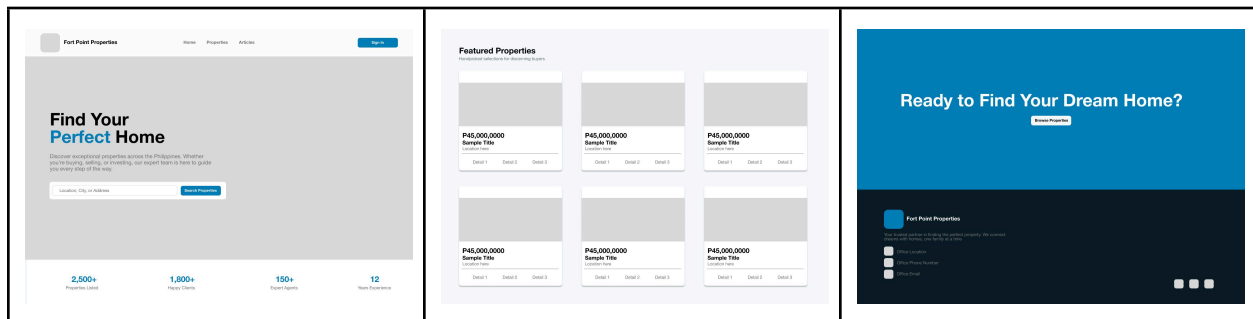
## 7.0 UI/UX DESIGN

### 7.1 Web Design Layout

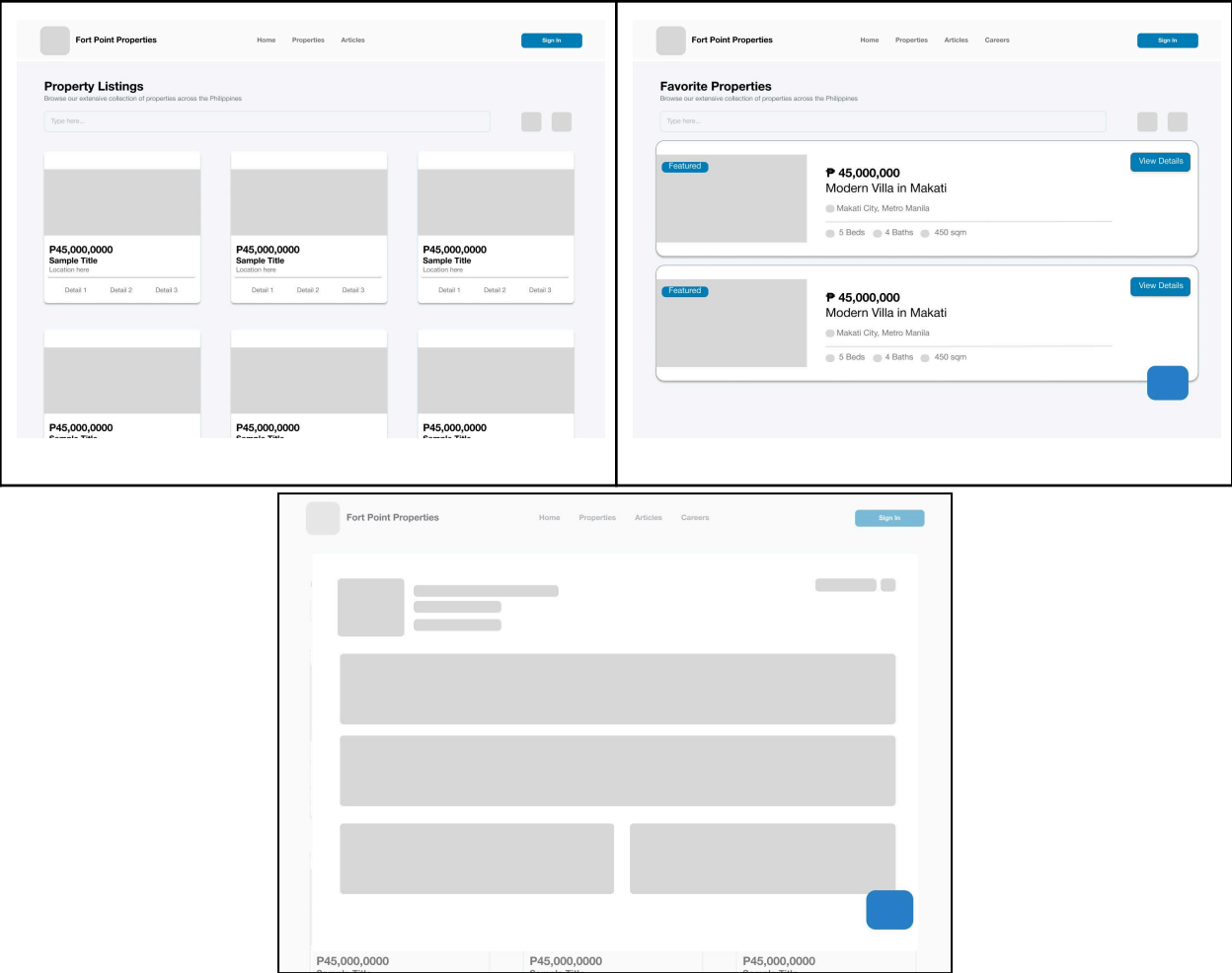
#### User Login & Registration



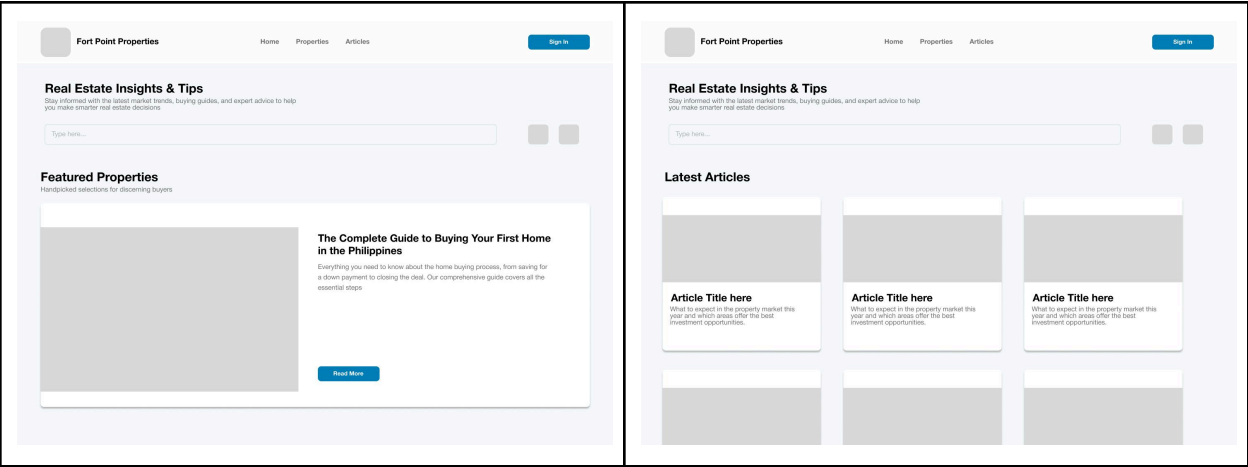
#### Home Page



#### Property Listing & Favorite Properties



## Articles

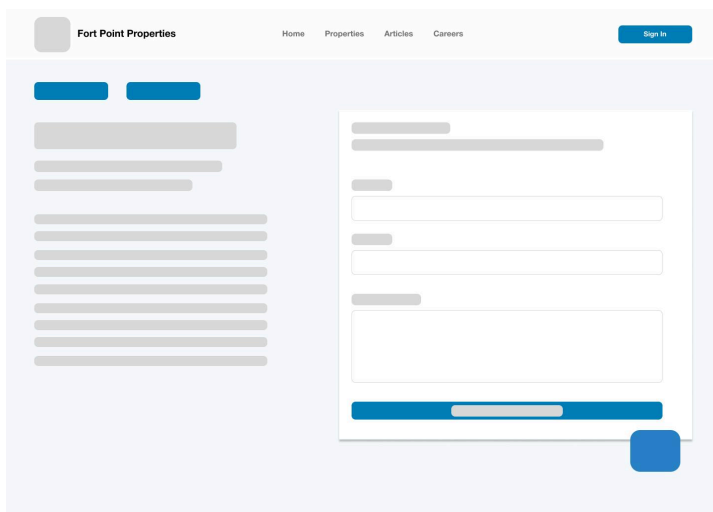


## In App Chatbox

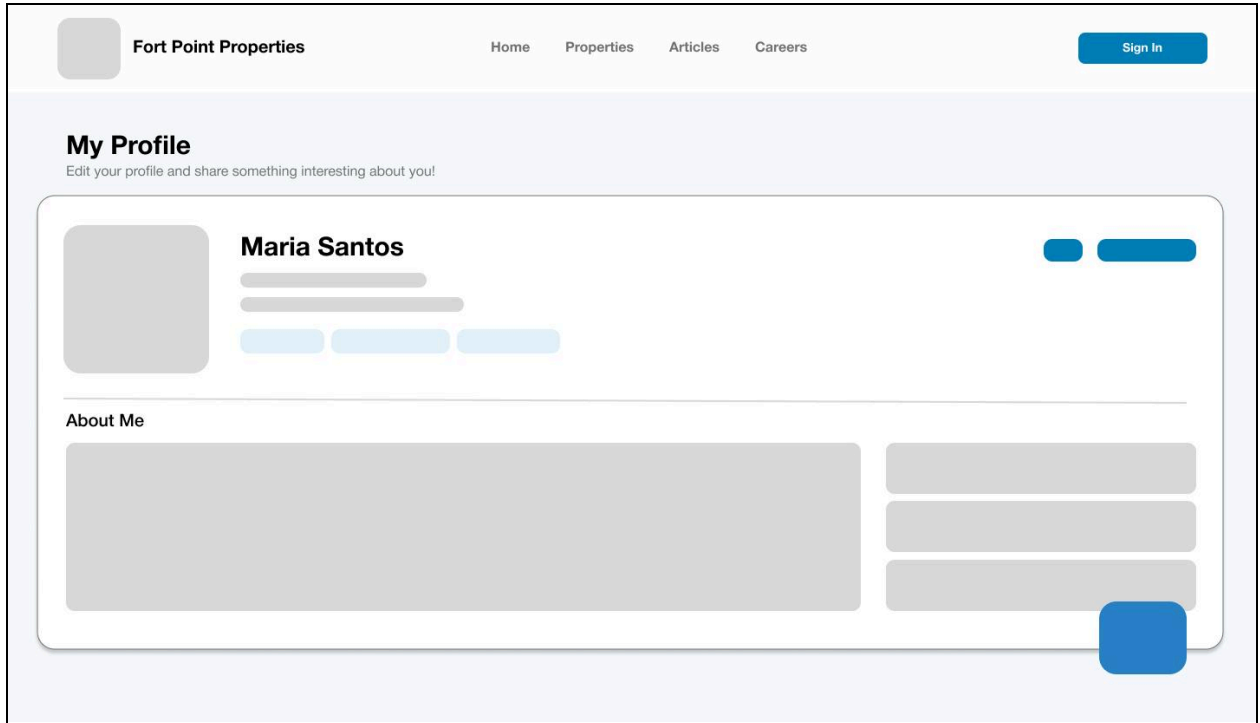




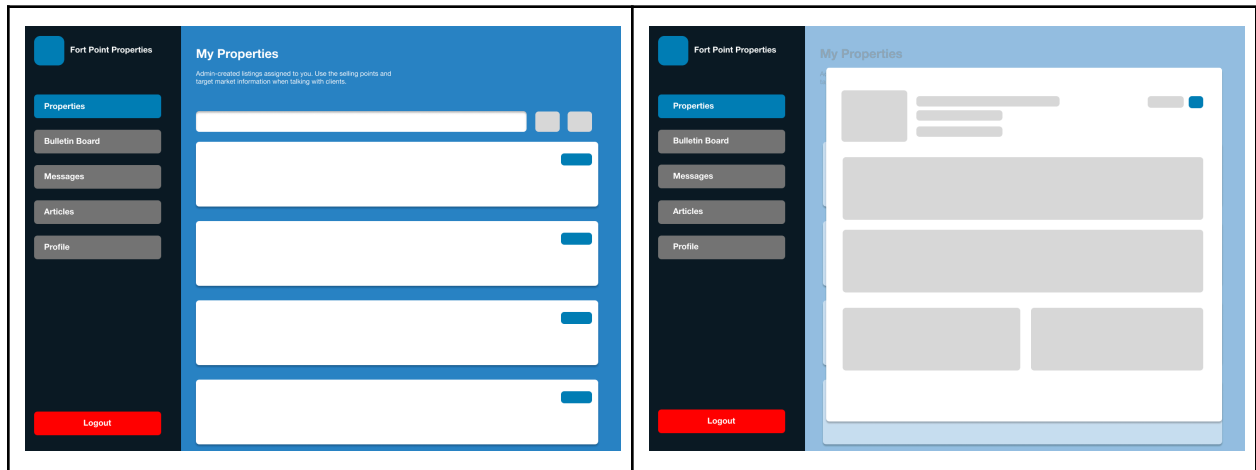
## User Job Application



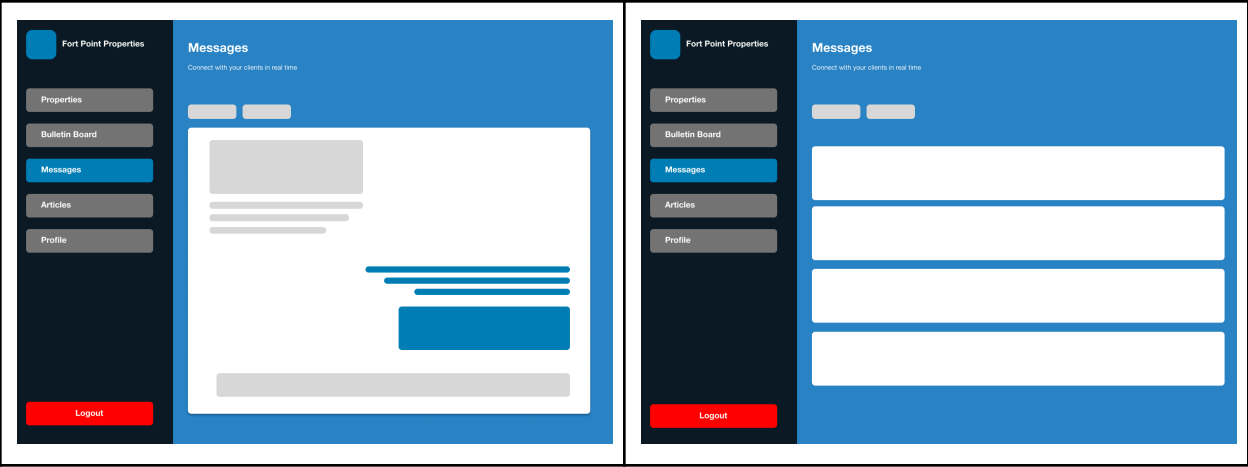
## User Profile



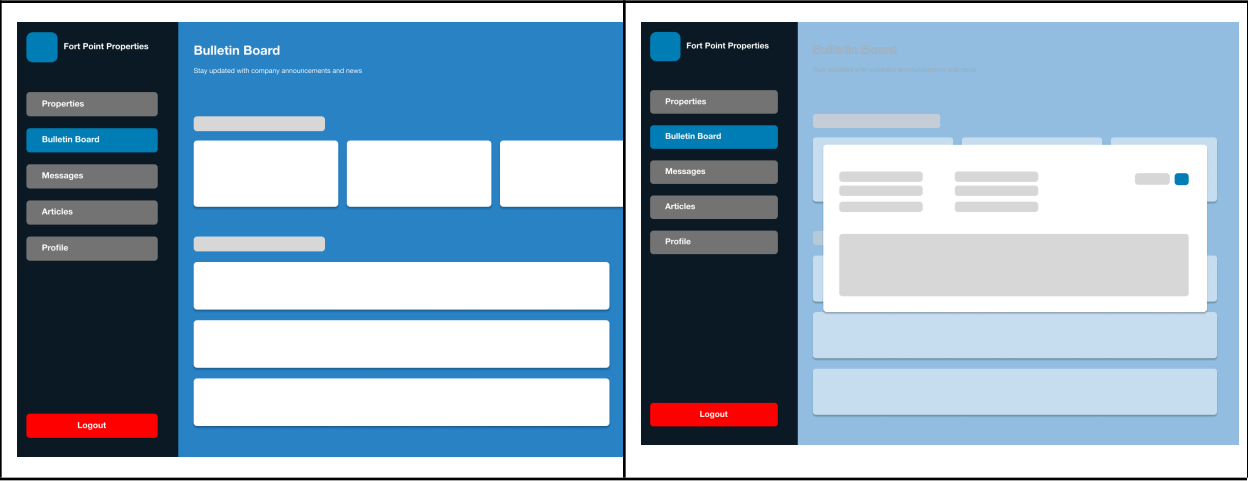
## Agent Properties



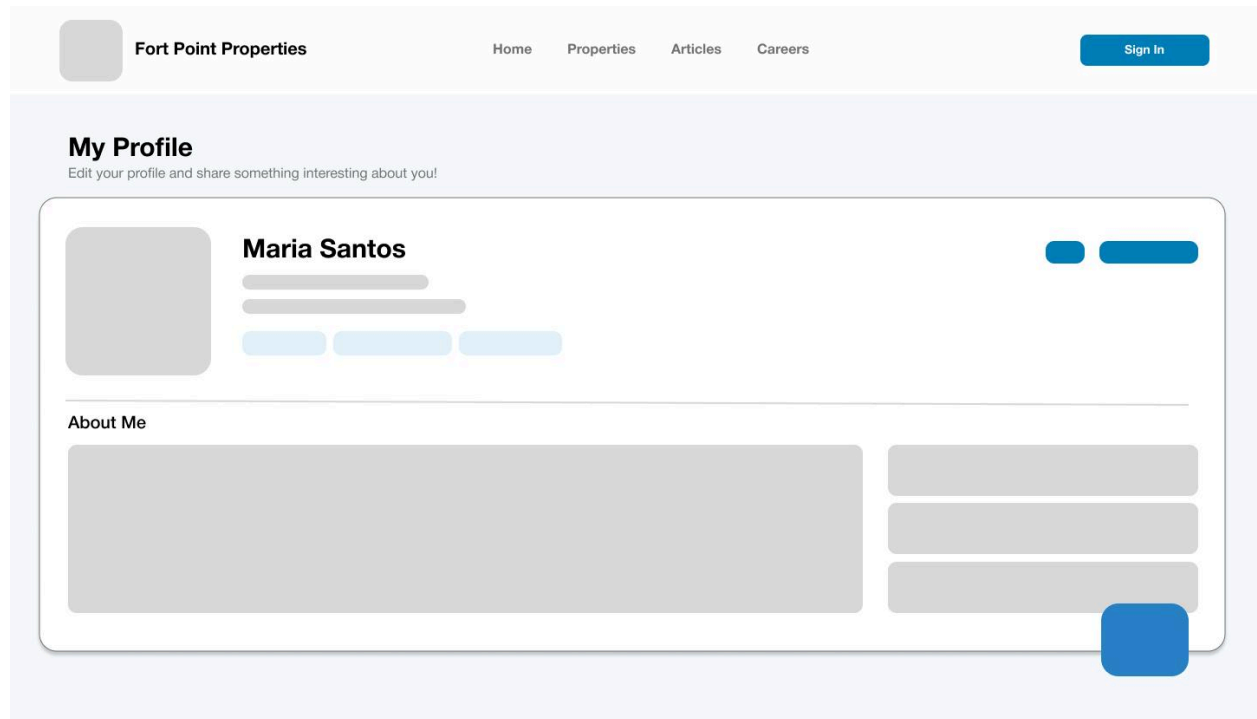
## Agent Client Conversation List



Agent Bulletin Board

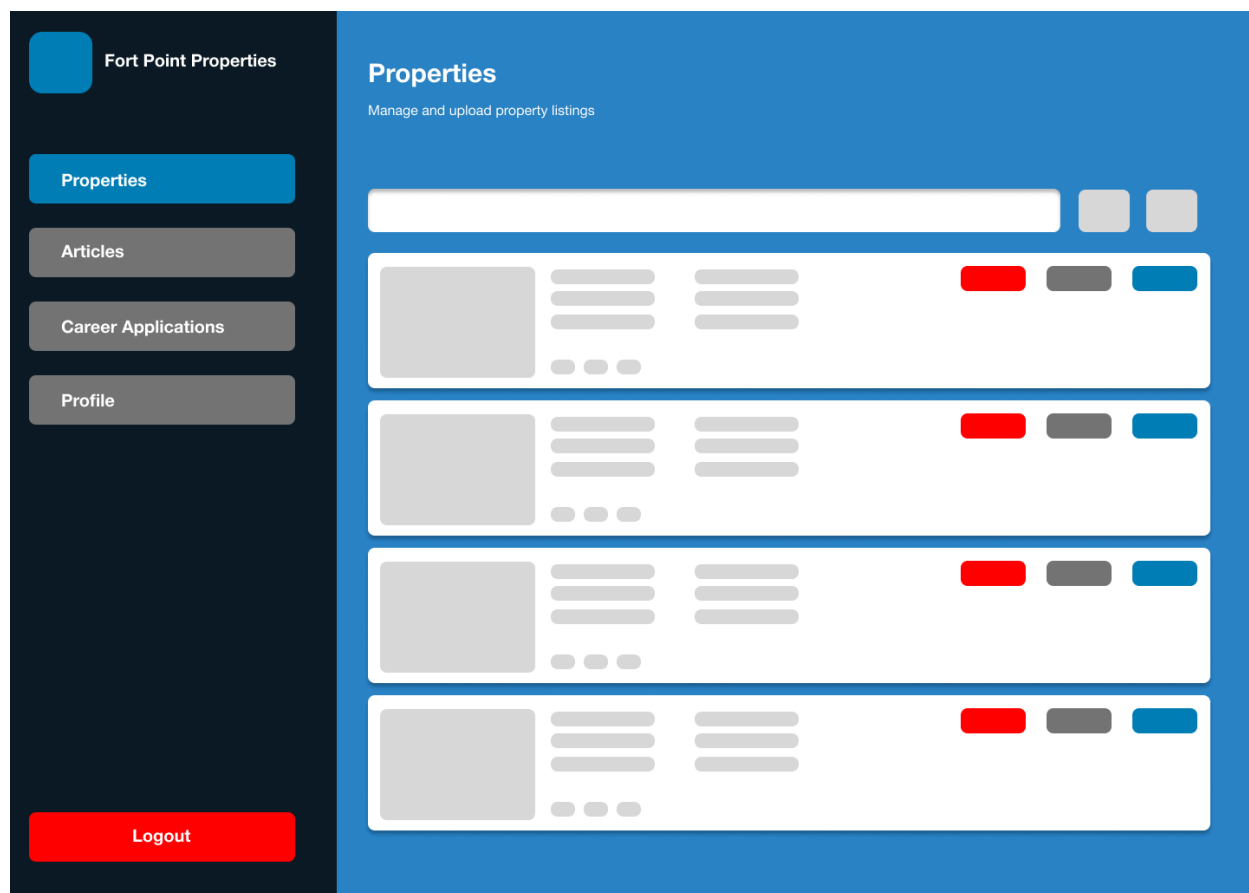


Agent Profile

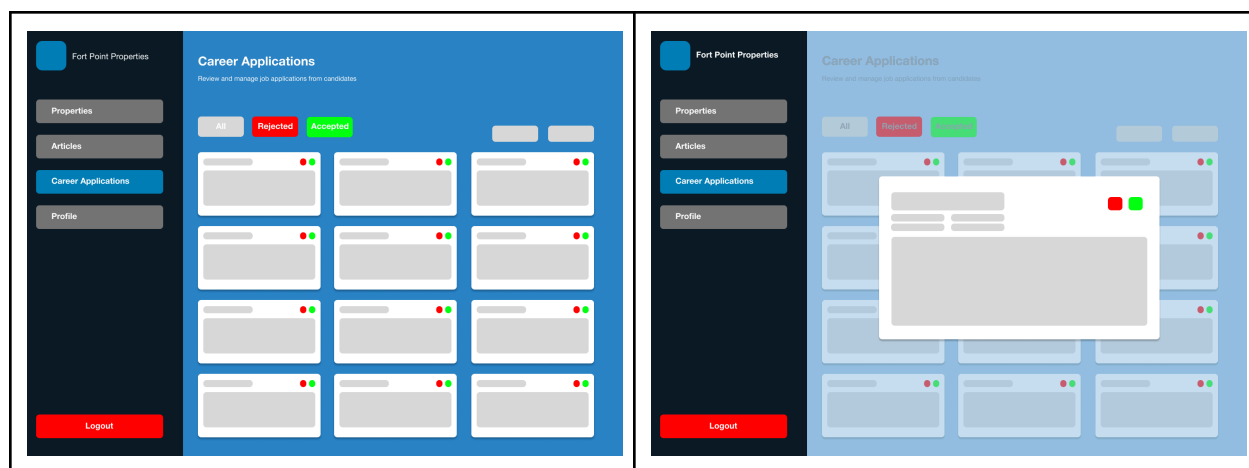


**Admin**

**Admin Property List**



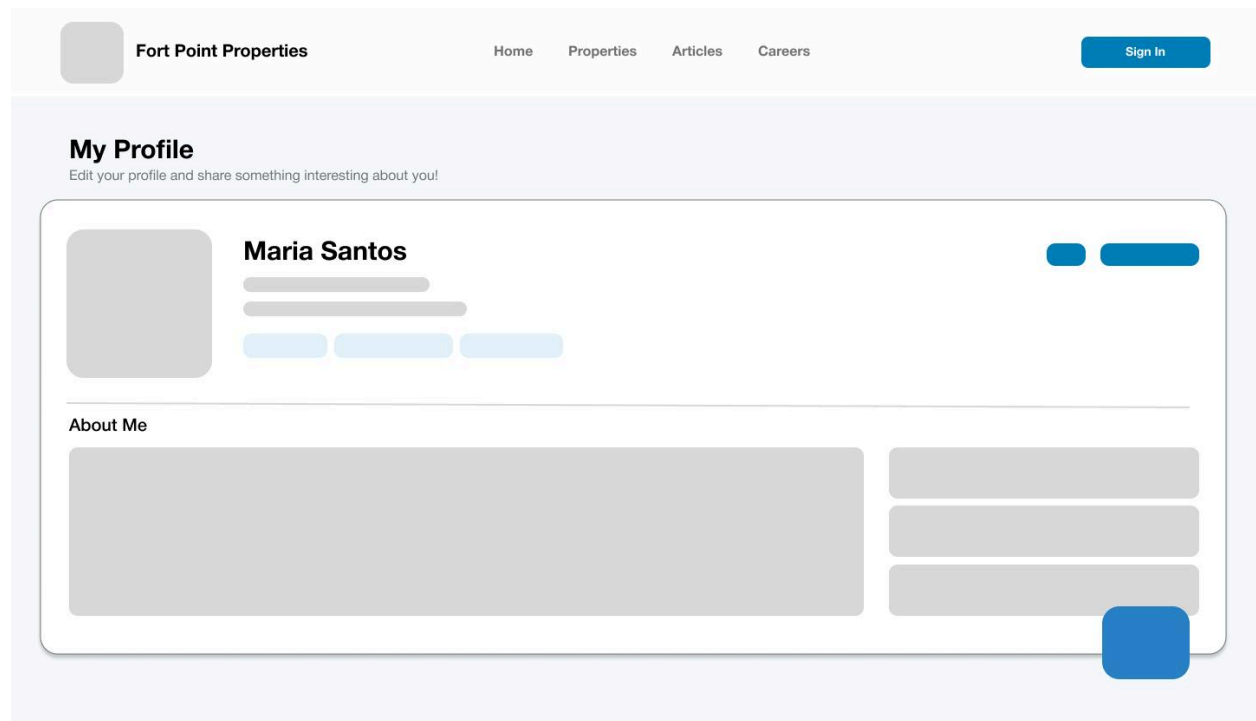
## Admin Job Applications



## Admin Articles



## Admin Profile



## 7.1 Mobile Design Layout

### User Registration

## Welcome!

Sign up to access to your account and manage your properties

First Name

Last Name

Email

Password

Confirm Password

Register

## Welcome!

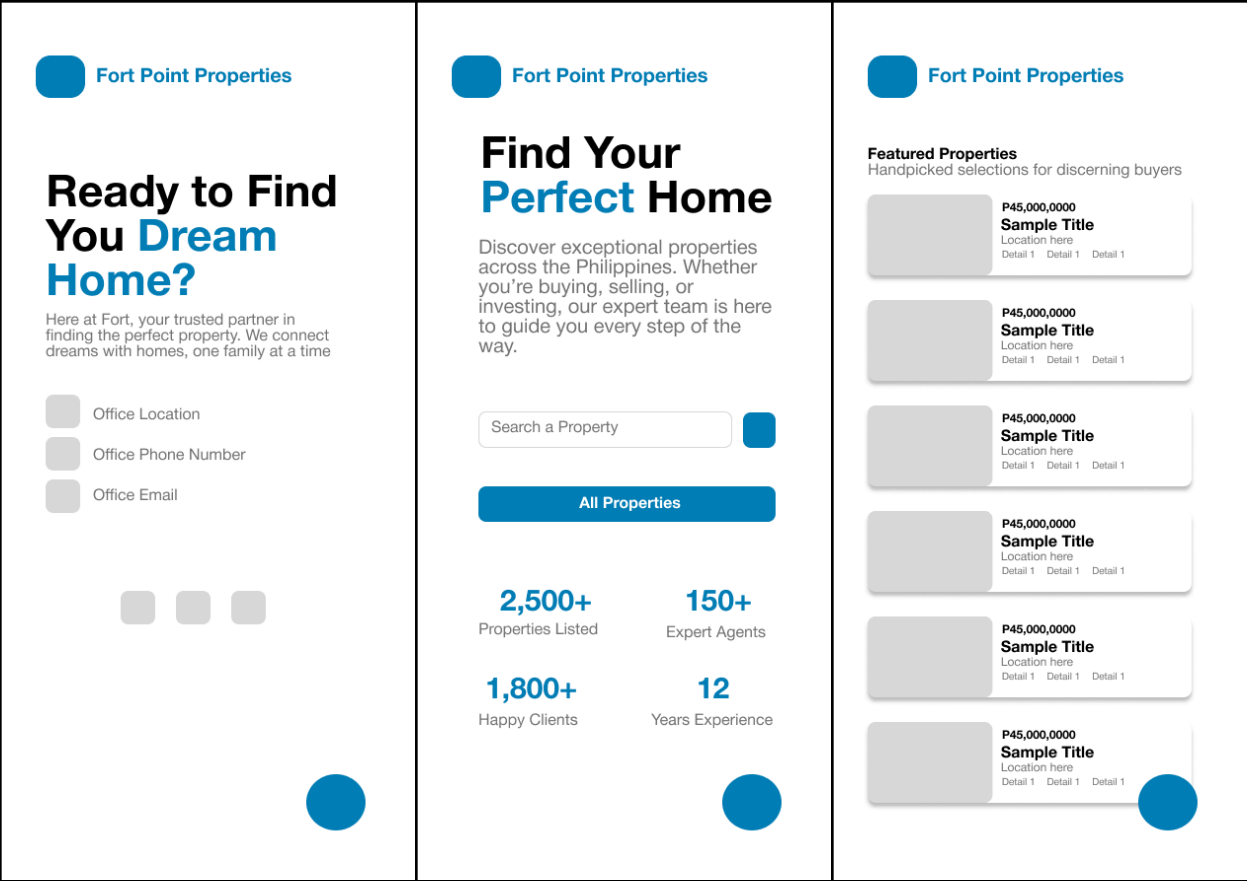
Sign in to access to your account and manage your properties

Email

Passowrd

Login

User Home Page



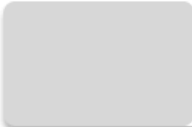
User Property Listing & Favorites





## Favorites

Browse our extensive collection of properties across the Philippines



P45,000,0000  
**Sample Title**

Location here

Detail 1 Detail 1 Detail 1



P45,000,0000  
**Sample Title**

Location here

Detail 1 Detail 1 Detail 1



P45,000,0000  
**Sample Title**

Location here

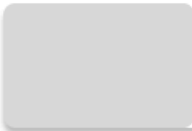
Detail 1 Detail 1 Detail 1



P45,000,0000  
**Sample Title**

Location here

Detail 1 Detail 1 Detail 1



P45,000,0000  
**Sample Title**

Location here

Detail 1 Detail 1 Detail 1



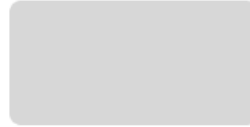
P45,000,0000  
**Sample Title**

Location here

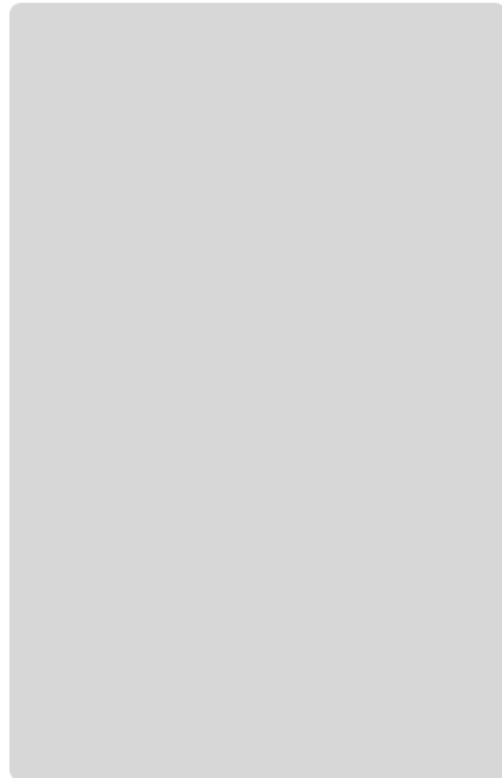
Detail 1 Detail 1 Detail 1



## Property Sample



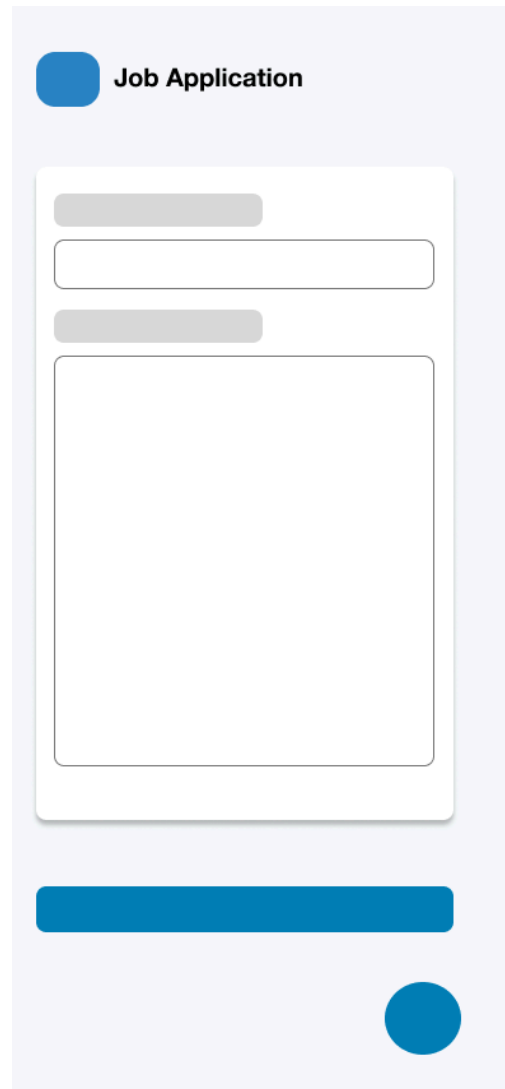
P45,000,0000  
**Sample Title**  
Location here



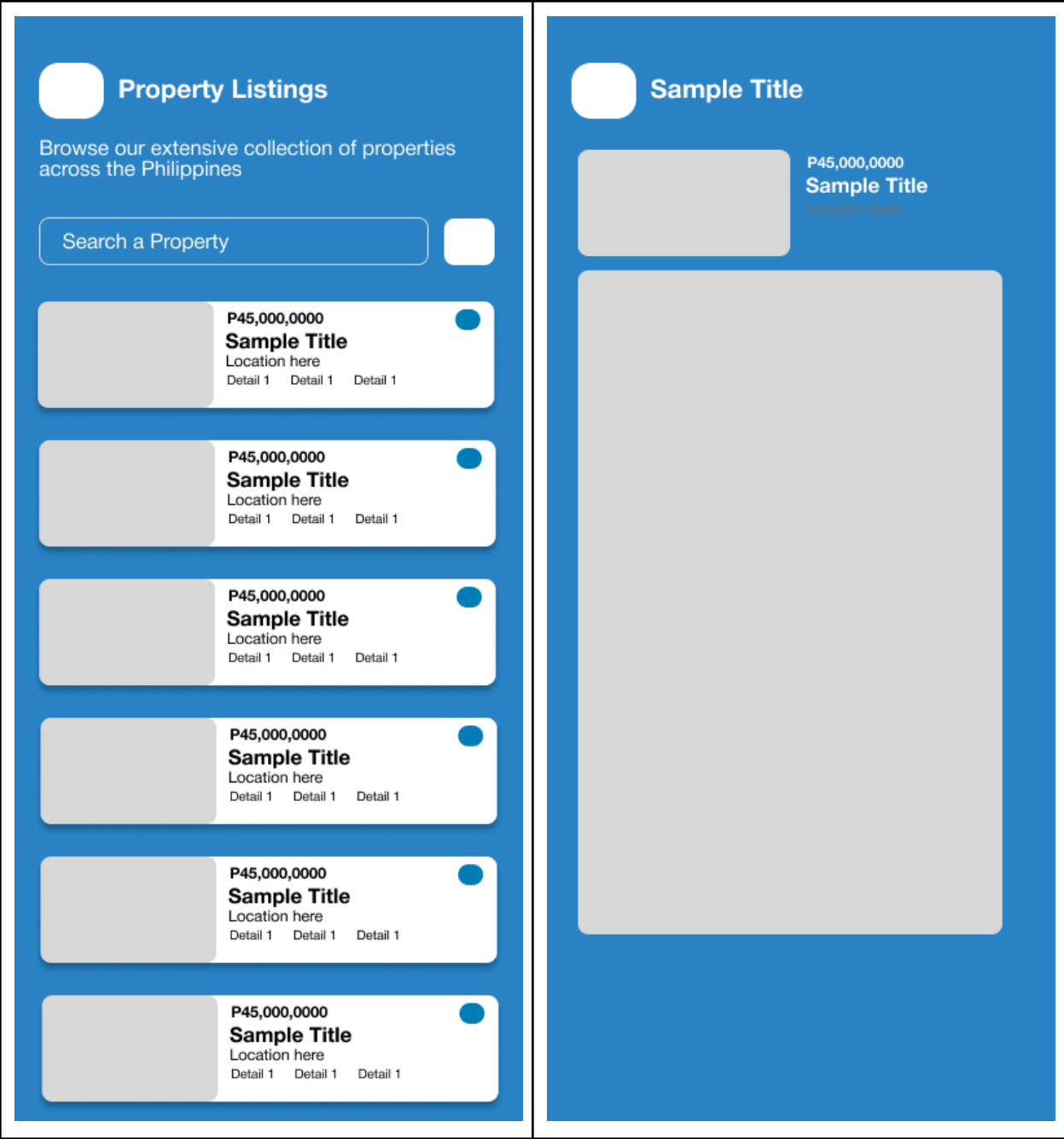
## In-App Messaging



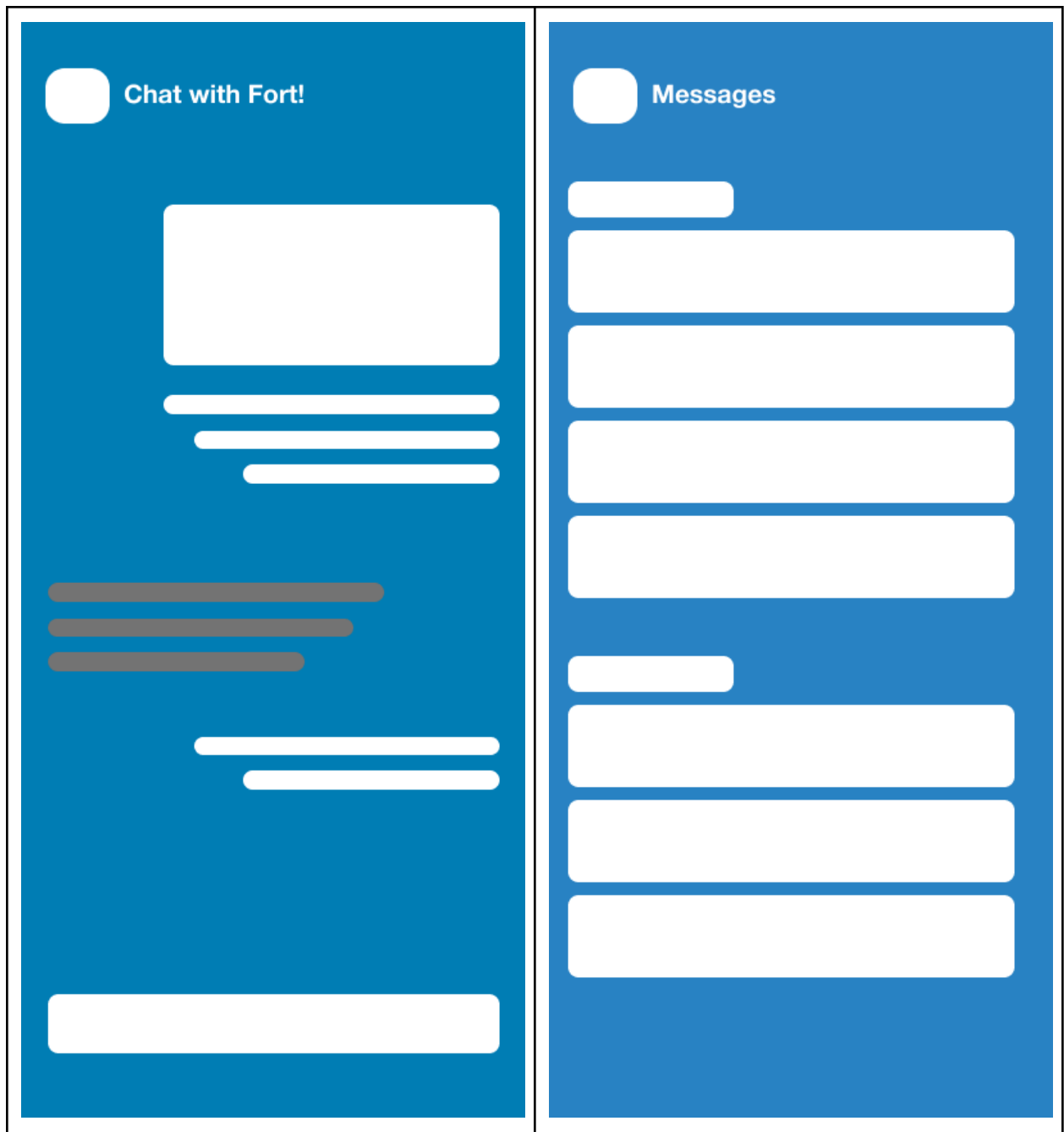
## User Job Application



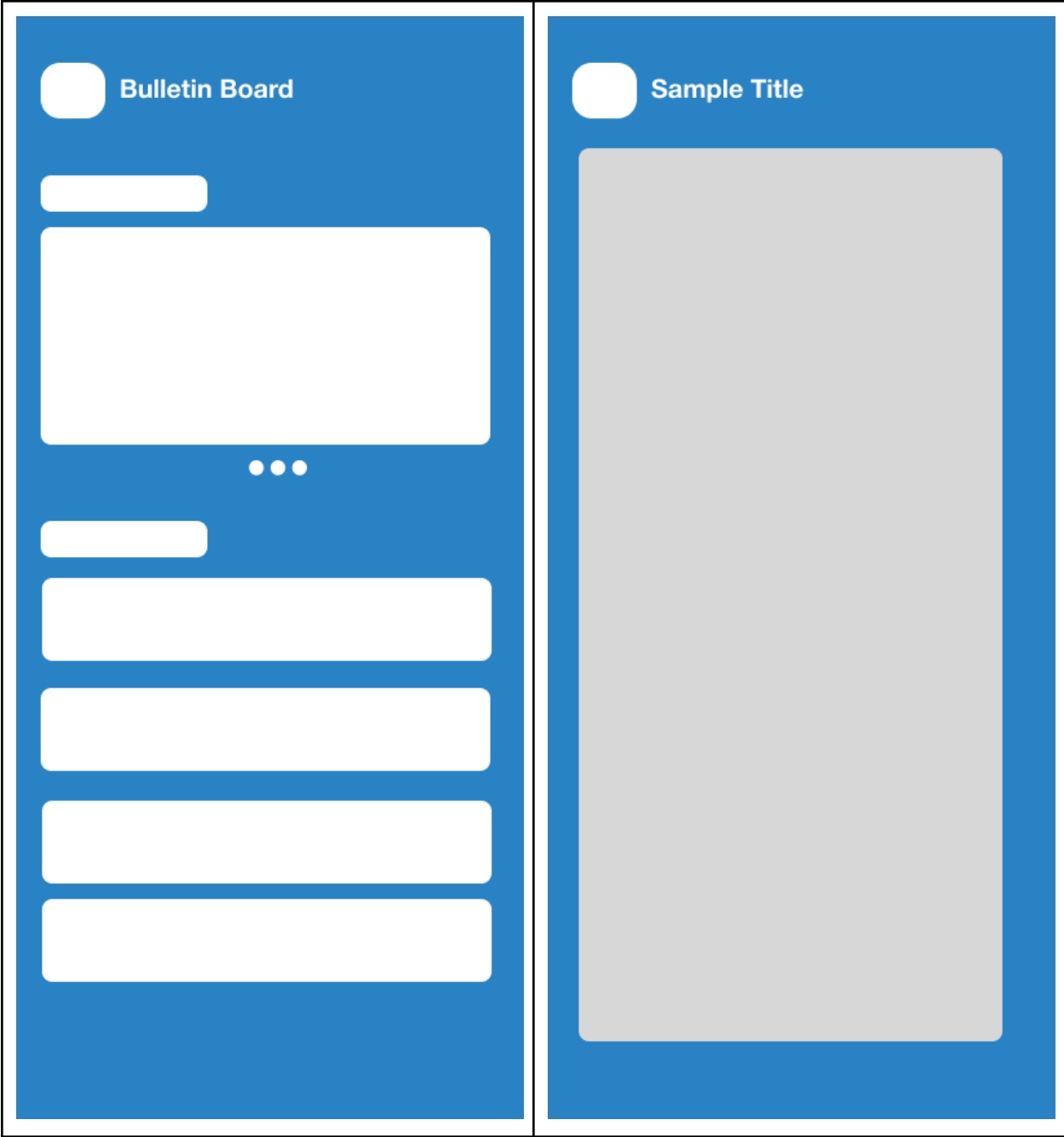
## Agent Property Listing



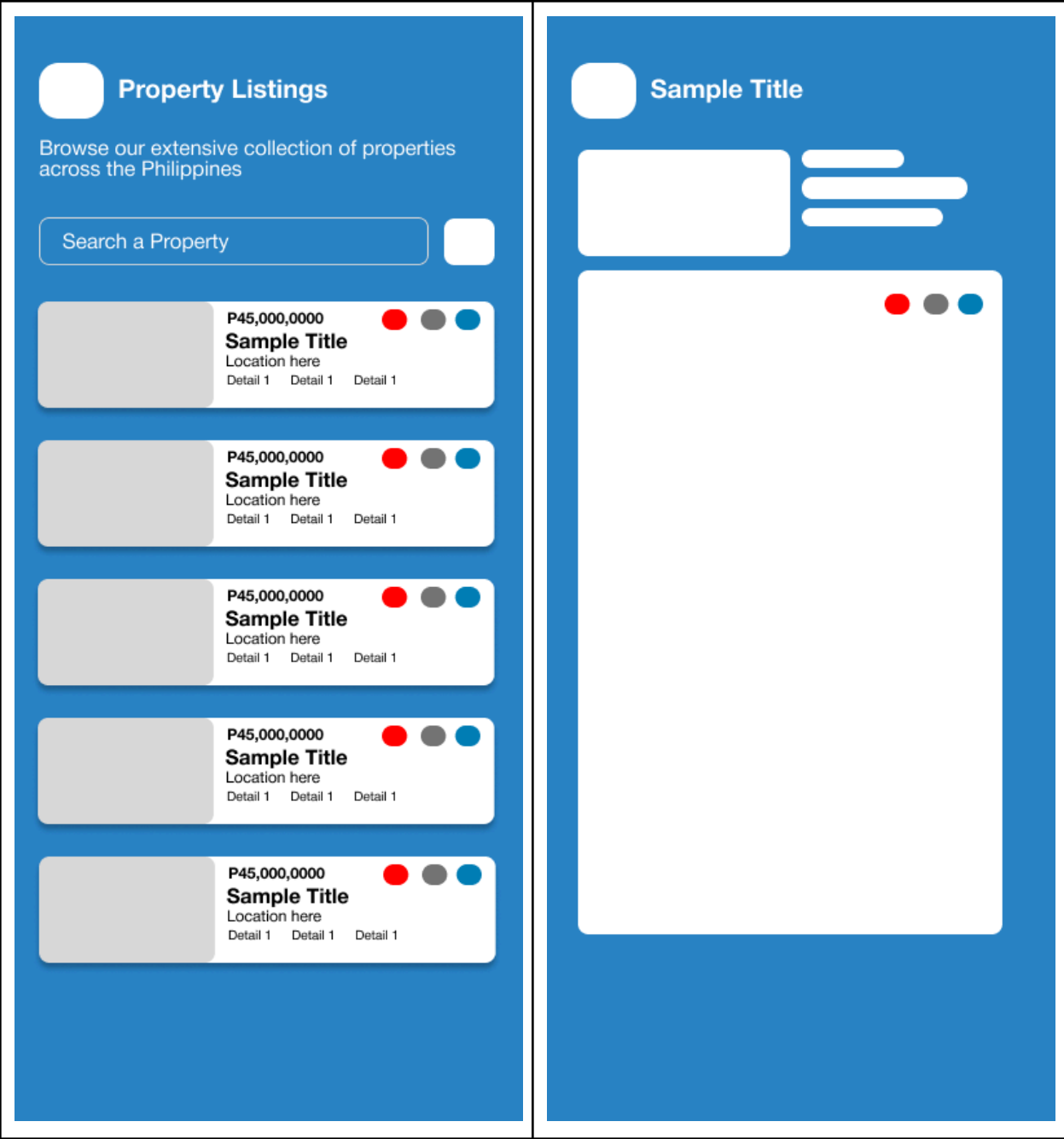
Agent In-App Messaging



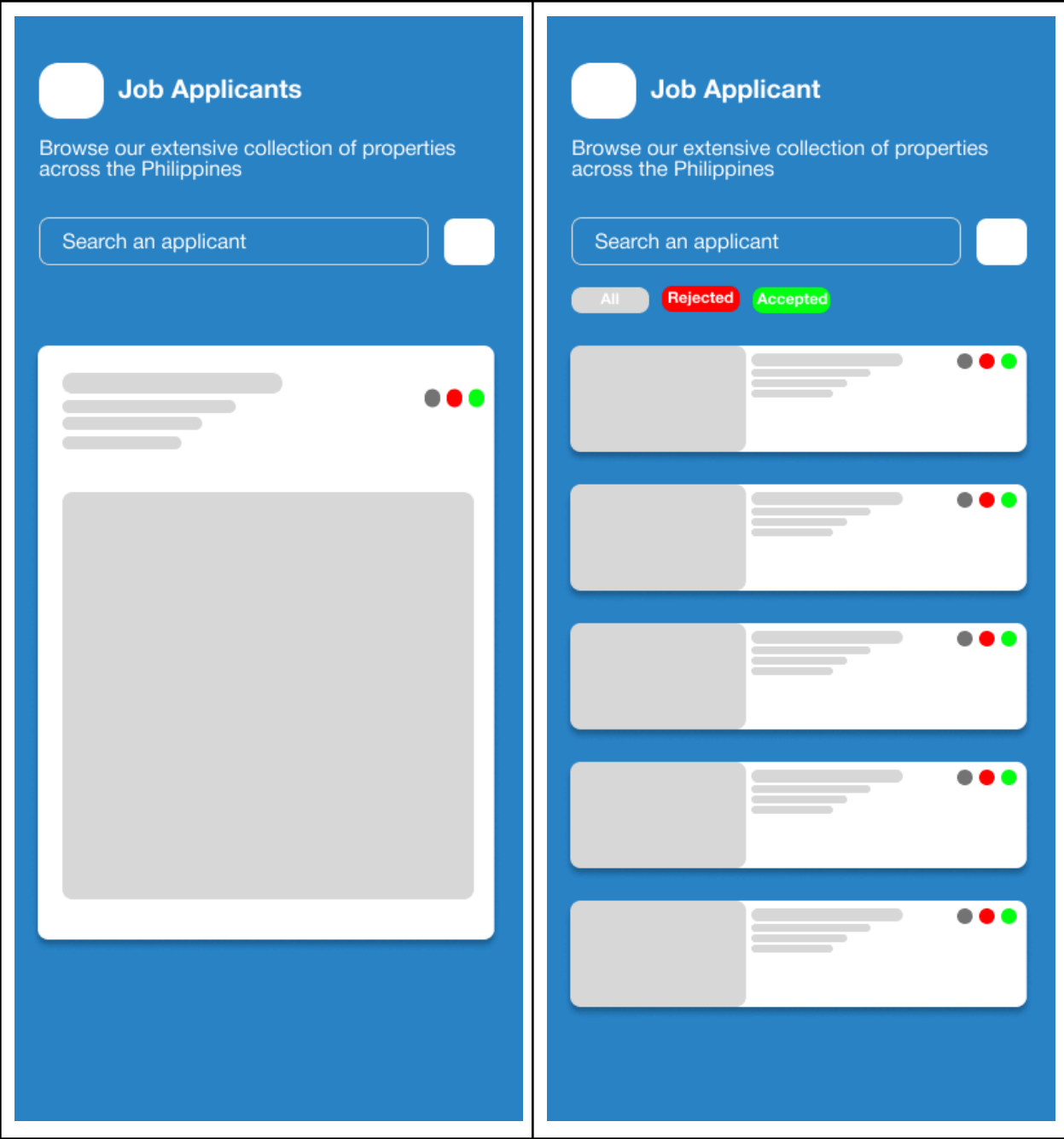
**Agent Bulletin Board**



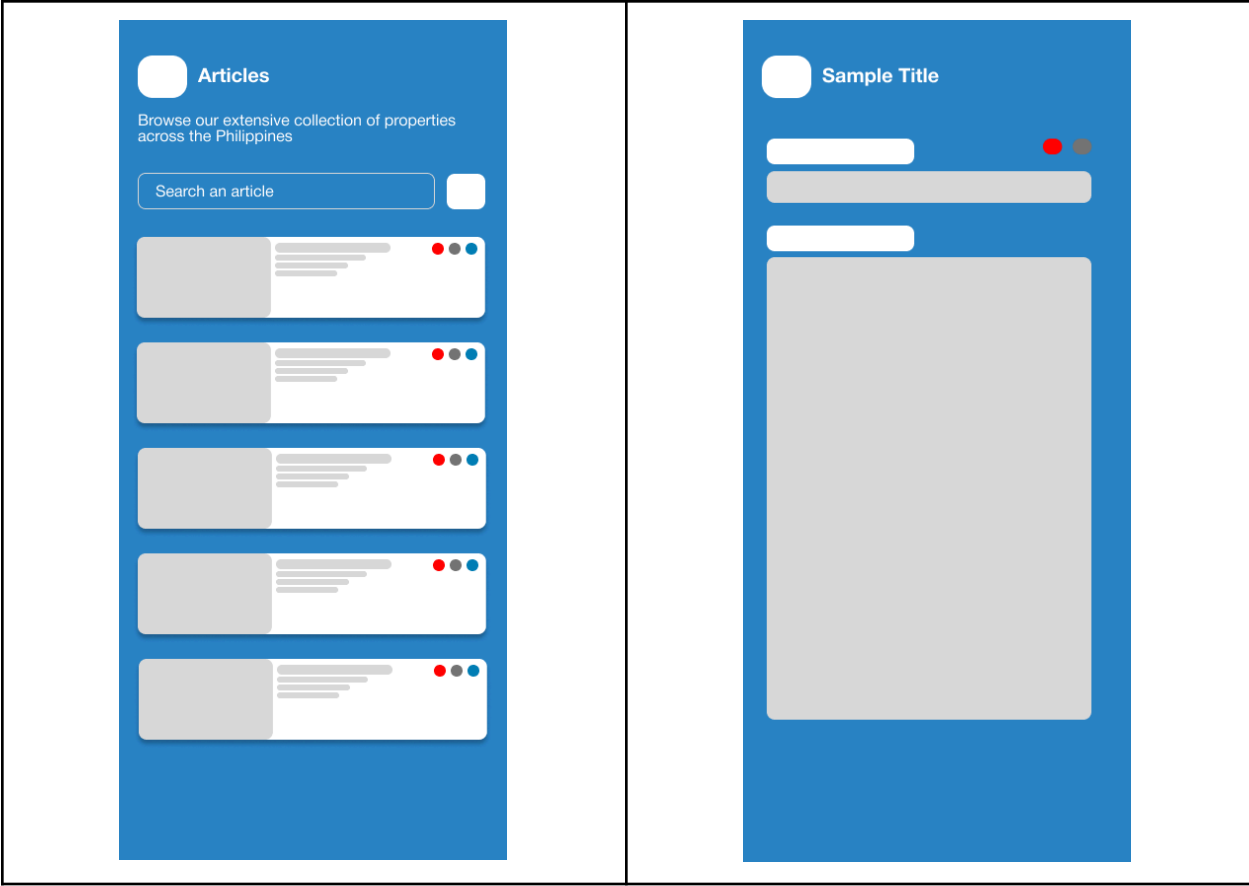
Admin Property Listing



**Admin Job Applications**

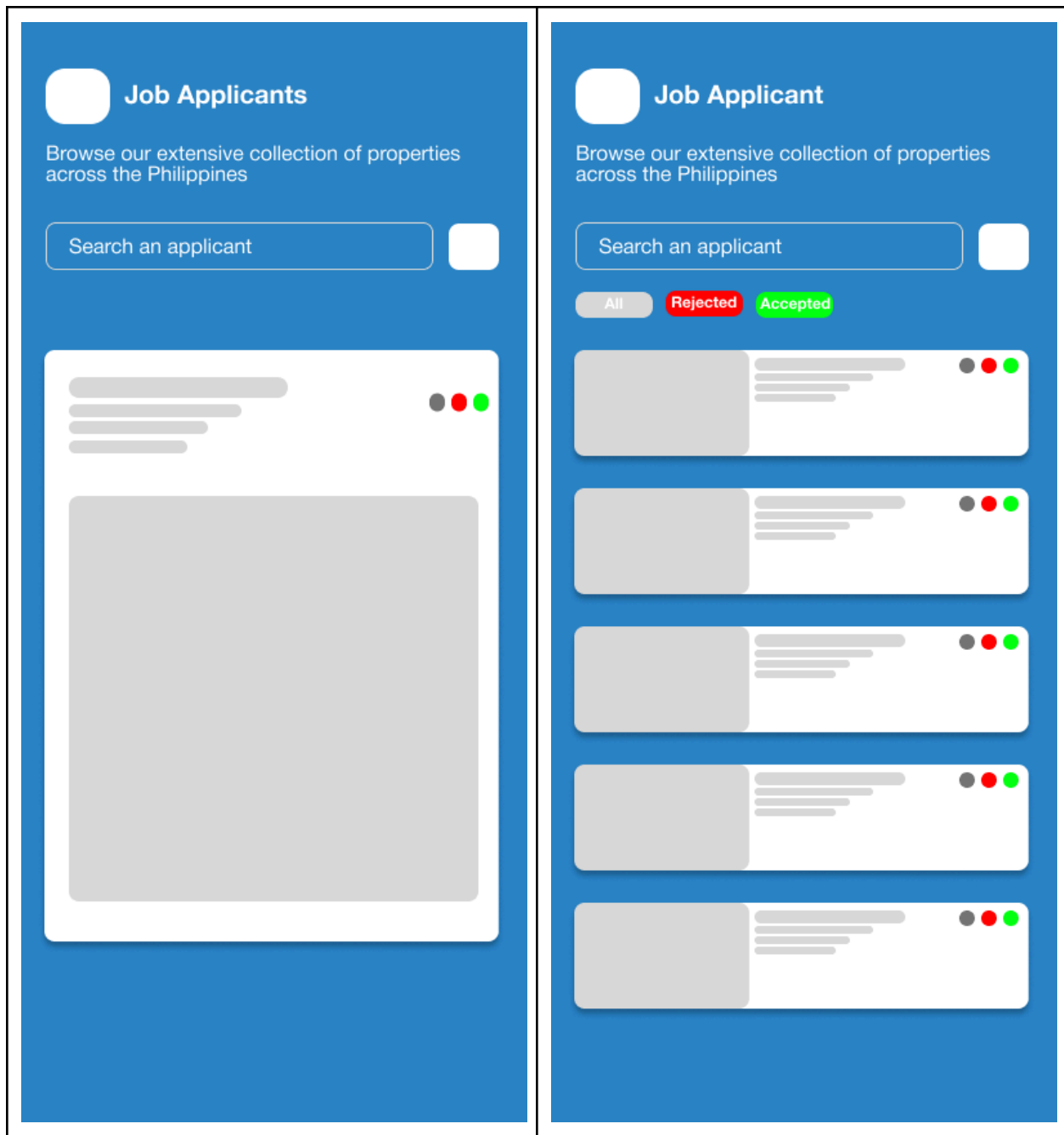


**Admin Articles**



**Admin Job Applications**





## 8.0 PLAN

### 8.1 Project Timeline

#### Phase 1: Planning & Design (Week 1-2)

- Week 1: Requirements & Architecture
  - Day 1-2: Finalize SDD and database schema.

- Day 3-4: Set up GitHub repository and basic Render hosting environments.
- Week 2: Detailed Design
  - Day 1-2: Map out UI/UX wireframes for the core pages.
  - Day 3-4: Define standard JSON structures for API requests.

## **Phase 2: Backend Development (Week 3-4)**

- Week 3: Foundation
  - Day 1-2: Spring Boot setup and MySQL database connection.
  - Day 3-4: Implement User Registration and basic JWT Login.
- Week 4: Core Features
  - Day 1-2: Build CRUD endpoints for Properties and the `property_units` table.
  - Day 3-4: Build messaging and favorites endpoints.

## **Phase 3: Web Application (Week 5-6)**

- Week 5: Frontend Foundation
  - Day 1-2: React setup, Tailwind configuration, and routing.
  - Day 3-4: Build Public Homepage and Login/Register pages.
- Week 6: Complete Web Features
  - Day 1-2: Build the Admin Dashboard to add/edit properties.
  - Day 3-4: Build the 3-column pricing table view for logged-in users.

## **Phase 4: Mobile Application (Week 7-8)**

- Week 7: Android Foundation
  - Day 1-2: Android Studio setup, XML/Compose UI for Login.
  - Day 3-4: Retrofit setup to consume the Spring Boot API.
- Week 8: Complete Mobile App
  - Day 1-2: Build property browsing and detail screens.
  - Day 3-4: Build the basic messaging interface.

## **Phase 5: Integration & Deployment (Week 9-10)**

- Week 9: Integration Testing
  - Day 1-2: Test Web and Mobile connecting to the live Render API.
  - Day 3-4: Fix bugs and clean up UI glitches.
- Week 10: Finalization
  - Day 1-2: Ensure Render database is stable and populated with sample CIT-area real estate data.

- Day 3-4: Record demonstration video and prepare for defense.

**Milestones:**

- M1 (End Week 2): Document and Figma designs approved.
- M2 (End Week 4): API deployed on Render and fully testable via Postman.
- M3 (End Week 6): React web app can successfully create and view properties.
- M4 (End Week 8): Android app can log in and view the property catalog.

**Critical Path:**

1. Database Schema -> 2. JWT Authentication -> 3. Property CRUD APIs -> 4. Admin Web Interface -> 5. Mobile Browsing Interface.

**Risk Mitigation:**

- **Time Constraints:** Focus strictly on the "MUST HAVE" features first. If time runs short due to other coursework, the Career Application portal ("SHOULD HAVE") can be dropped.
- **Deployment Issues:** Deploy the backend to Render in Week 3, not Week 10, to catch hosting errors early.